Erin Fels

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Education

University of Colorado Boulder, College of Engineering and Applied Science *Expected Graduation: May 2028*

Major: (B.A.) Computer Science GPA: 4.0

Major: (B.S.) Creative Technology and Design **Minor:** Business

Relevant Coursework: Human-Computer Interaction, Graphic Design, Design a Science Museum Exhibit, Image, Physical Computing, Software Development, Computer Systems, Algorithms, Data Structures, Mechatronics, Intro to Robotics

Societies & Affiliations: BOLD Scholar, Tau Beta Pi, Society of Women Engineers, CU Women in Computing

Skills and Qualifications

Programming: C++, JavaScript, HTML, CSS, SQL, C, Assembly, MQL, CQL, Arduino IDE, API Testing (Postman), GitHub Version Control, PlatformIO, VS Code

Design Software: Adobe Creative Suite, Figma, Fusion 360, Pangolin Beyond, GrandMA, Verge Design Studio, Ableton

Fabrication: Soldering, 3D Printing, Laser Cutting, Woodworking

Work Experience

Design and Programming Intern, Image Engineering **June 2026 - August 2026**

- Designed special effect shows for events, utilizing lasers, drones, lighting, and pyrotechnics for multimedia effects
- Set up and teched special effects equipment (lasers, lighting, pyro), ensuring live events are run successfully and safely
- Participated in the creative planning and communicated with clients to deliver their expected vision

Undergraduate Researcher, Unstable Design Lab **January 2025 - May 2025**

- Created a prototype interactive device to allow hands-on control of E-Textiles and observe human-computer interaction
- Designed and soldered circuitry safely integrating an Arduino with a motor, LEDs, and a MIDI controller
- Coded in C++ using Arduino IDE to control the motor, LED responses, and digital and analog inputs

Learning Assistant, CU Boulder Atlas Institute **December 2025 - Present**

- Assisted an average of 12 students weekly with topics on physical computation and product design
- Taught recitations, graded homework/projects, held office hours, and provided feedback to enhance understanding
- Instructed students in circuitry, coding, 3D printing, and laser cutting to complete their projects and homework

Engineering & Project Experience

Vice President, CU Theme Park Engineering and Design Club **April 2026 - Present**

Electrical Theming Lead, CU Theme Park Engineering and Design Club **April 2025 - April 2026**

- Created a design scheme and style guide for the ride to create a cohesive and appealing ride
- Worked with a team of 5 to research and select a theme to create a marketable and entertaining experience
- Specced electrical components that enhanced the overall theme
- Designed ride logo and summary poster, using Illustrator, helping the team place 2nd overall and in theming out of 16 teams
- Created technical graphics outlining a state system and control panel for design reviews and documentation

Front-End Software Engineer, Software Development Method and Tools **February 2026 - May 2026**

- Worked with a team of 5 to replace and improve the UI/UX and accuracy of the CU Dining Hall menu website
- Engineered and documented a full-stack software design website using HTML, CSS, Bootstrap, and JavaScript
- Created a style guide with a color palette, fonts, graphics, and a logo to facilitate the creation of a cohesive design

Project Manager, ATLAS Light Tower as a Theater **January 2026 - May 2026**

- Revamped an unused Light Tower to make a multimedia workspace for CU Students
- Coordinated with faculty to secure funding for new equipment (lights, speakers, lasers) to upgrade the existing space
- Led a team of 9 master's and undergraduate students ranging from artists to engineers to create an immersive lighting show
- Documented the project for others to make their own Light Tower shows

Technology Lead, ATLAS How To: Design a Science Museum Exhibit **January 2026 - May 2026**

- Worked with a team of 6 to create an interactive exhibit for users of all ages to educate them about the aurora borealis
- Integrated a website with physical interfaces via an ESP-based MCU, Wi-Fi, and WebSockets for real-time interaction
- Designed and soldered the circuitry and control system to create safe and reliable connections
- Created a printable take-home token for users by integrating a physical interface and a printing system using Node.js
- Documented the process, code, materials, and “recipe” for others to repeat and recreate for themselves

Social Media Lead, CU Theme Park Engineering and Design Club **April 2025 - May 2026**

- Managed all social media and public-facing accounts, including LinkedIn, Instagram, and TikTok
- Created and designed graphics, posters, videos, and photos that would catch attention to increase awareness and attendance